

RaahSetu — Frontend Architecture & Technology Report

Comprehensive Technical Blueprint & End-to-End Implementation of the Risk-Aware Logistics Routing System for Northeast India

DOCUMENT ID	PLATFORM ACCESS	REGIONAL DOMAIN	ARCHITECTURE MODE
RS-TECH-FE-2026	http://127.0.0.1:8000/	8 Northeast States (111 Nodes)	Single-Port Unified Delivery

1. Executive Summary & Operational Context

RaahSetu is an enterprise logistics navigation and corridor accessibility intelligence platform designed for the extreme terrain and climate vulnerabilities of Northeast India (Assam, Arunachal Pradesh, Meghalaya, Manipur, Mizoram, Nagaland, Tripura, and Sikkim). Conventional commercial mapping systems (such as Google Maps or Mapbox standard navigation) optimize strictly for linear distance or typical passenger-car traffic congestion. In mountain ghat corridors, this naive shortest-path approach routinely misroutes heavy multi-axle freight carriers into fatal 14%+ inclines causing brake fade, load-limited Bailey bridges (<12 Tonnes), and active landslide zones (such as NH-10 Teesta Bazaar, NH-6 Sonapur tunnel, and Sela Pass).

The RaahSetu frontend operates as an **Offline-First, Dual-Solve Routing Client**. It pairs a high-performance local client-side Dijkstra algorithm (for instantaneous sub-millisecond calculation and offline mountain continuity) with an asynchronous FastAPI A* routing engine that queries real-time hazard registries, road elevation gradients, and PostGIS infrastructure closures.

Key Architectural Breakthrough: The entire frontend client (HTML, CSS, WebGL 3D meshes, Leaflet raster maps, and voice recognition modules) is compiled into a single static production artifact (`dist/`) and served directly from the root endpoint of the FastAPI engine on port `8000` . This provides zero-CORS communication, unified deployment, and instantaneous sub-millisecond API response.

2. Complete Technology Stack Matrix

Technology / Library	Version	Functional Role	Technical Implementation & Architectural Rationale
React	19.1.0	Core Reactive UI Framework	Provides declarative UI rendering, concurrent transitions, automatic render batching, and strict functional component lifecycles (<code>useMemo</code> , <code>useCallback</code> , <code>useRef</code>).
TypeScript	~5.9.2	Static Type Safety & Verification	Guarantees compile-time integrity across 100+ interfaces (e.g. <code>DriverProfile</code> , <code>Comparison</code> , <code>RouteLeg</code> , <code>Translations</code>) preventing runtime failures during emergency freight dispatch.
Vite	7.1.0	Build System & Modern Bundler	Provides native ES module HMR in development and Rollup-powered tree-shaking and minification in production, outputting optimized <code>dist/assets/</code> bundles.
Tailwind CSS	4.3.3	Next-Gen Utility Styling Engine	Tailwind CSS v4 introduces engine-level compilation via <code>@tailwindcss/vite</code> , OKLCH high-gamut color spaces, container queries, and seamless CSS custom variable dark/light theming.
Leaflet	1.9.4	2D Geospatial Map Canvas	Renders OpenStreetMap raster tiles, custom SVG pinpoint markers, hazard overlay rings, and dual polyline paths (fastest linear vs. risk-aware recommended corridor).
Three.js	0.180.0	3D WebGL Graphics Pipeline	Low-level WebGL graphics engine rendering custom Digital Elevation Models (DEM), procedural heightmaps, and vertex shader gradient coloration for mountain terrain.
React Three Fiber	9.3.0	Declarative Three.js Bridge	Bridges Three.js scene graphs directly into React's virtual DOM reconciliation loop, allowing declarative lights, cameras, and meshes without memory leaks.
@react-three/drei	10.7.0	3D Scene & Camera Helpers	Supplies smooth <code>OrbitControls</code> , perspective camera rigs, and environmental lighting for 360-degree mountain terrain inspection.
Framer Motion	13.2.0	Fluid Micro-Interactions	Hardware-accelerated animations for route calculation updates, sliding drawers, animated stat counters, and modal dialogs.
Lucide React	0.468.0	Vector Iconography	Tree-shakeable SVG icons for mountain logistics (<code>Truck</code> , <code>Navigation</code> , <code>ShieldAlert</code> , <code>Scale</code> , <code>Gauge</code>).
Web Speech API	Browser Native	Speech-to-Text & TTS Voice Audio	Hands-free speech recognition (STT) for driver city search and text-to-speech (TTS) multilingual audio turn-by-turn navigation across 4 regional languages.
HTML5 Canvas API	Browser Native	Procedural Bot Defense Captcha	Generates random alphanumeric verification codes with character jitter and noise curves directly on client canvas, eliminating tracking cookies and third-party dependencies.

Technology / Library	Version	Functional Role	Technical Implementation & Architectural Rationale
Fetch & AbortController	Browser Native	Type-Safe REST Client	Executes asynchronous HTTP POST/GET requests to FastAPI backend with automatic query cancellation on origin/destination re-selection.
HTML5 LocalStorage	Browser Native	Offline Persistence & Queue	Persists language mode, active driver specifications, SOS distress logs, and queued offline hazard reports during high-pass signal blackouts.

3. End-to-End Architectural Subsystem Breakdown

3.1. Core Application Shell & State Controller (`App.tsx`)

`App.tsx` is the central nerve center of the RaahSetu user interface. It manages:

- **Discriminated Navigation Routing:** Implements a clean client-side state machine without bulky third-party router bloat: `activeView: "intro" | "auth" | "vehicle_details" | "dispatcher" | "districts" | "advisories" | "system"`.
- **Dual Theme Architecture:** Light mode is deliberately configured as the default for high-contrast visibility inside brightly-lit truck driver cabins during daytime mountain runs, while dark mode provides tactical low-glare visibility for nighttime dispatch control rooms.
- **Active Highway Closure Telemetry Marquee:** Cycles through critical real-world mountain road blockages (e.g. NH-10 Teesta Bazaar rockfall, NH-6 Sonapur tunnel mudslide, Sela Pass snow blocks) with instant 1-click inspection.
- **Live Backend Latency Telemetry:** Calculates round-trip API compute time in real time and renders an on-screen status pill (`• FastAPI A* Connected (XXms)`).

3.2. 2D Interactive Map

Engine (

`RealMapLeaflet.tsx`)

Built on Leaflet 1.9.4 inside a React `useRef` container. Renders OpenStreetMap raster tiles with dynamic retina scaling. Features a 4-tier layer hierarchy:

- **Baseline Corridor Network:** 378 regional highway links in slate blue.
- **Fastest Route:** Amber dashed polyline showing linear distance.
- **Risk-Aware Route:** Vibrant emerald polyline with lower risk exposure.
- **Hazard Pins:** Pulsating alert markers with contextual roadblock details.

3.3. 3D Terrain Elevation

Engine (

`TerrainMap.tsx`)

Leverages Three.js, React Three Fiber (R3F), and Drei. Renders Digital Elevation Models (DEM) from `/api/v1/terrain` into a 3D elevation mesh.

- **Procedural Gradient Shader:** Color-codes altitude from Brahmaputra river valleys (green) to Himalayan peaks (white snow).
- **Full Orbit Controls:** Allows controllers to pitch, rotate, and zoom to verify incline gradients and cliff drops before dispatching heavy freight.

3.4. Authentication & Security (`AuthPage.tsx`)

Two-step credential and verification system:

- **Dual Mode:** Sign In for fleet drivers or Sign Up for new logistics operators.
- **Procedural Canvas Captcha:** Generates 6-character alphanumeric challenges with rotational distortion and distractor lines.
- **Two-Factor Email OTP:** Interfaces with backend `/api/v1/auth/request-code` with a 10-minute expiry countdown timer.

3.5. Post-Login

Vehicle Specs (

`VehicleDetailsPage.tsx`)

Appears immediately after successful login (Step 2 of 2):

- **1-Click Presets:** Heavy 12-Wheeler (16.5T), Medical (5.2T), Fuel Tanker (14.2T), 4x4 Bolero Pickup (2.8T), Agro Produce (9.5T).
- **Bailey Bridge Load Check:** Interactive GVW slider evaluates weight against Class 18/24 load ratings.
- **Tunnel Clearances:** Validates height/width against Sonapur (4.5m) and Sela (5.2m) portals.

3.6. Voice Navigation

Engine (

`voiceRecognition.ts`)

Hands-free driving safety via browser-native Web Speech API:

- **Phonetic Fuzzy Matching:** Soundex and Levenshtein algorithms map regional accents (e.g. "Guhatee", "Shilang") to standard node IDs.
- **Multilingual TTS Audio:** Speaks turn-by-turn guidance in English, Hindi, Assamese, or Bengali.

3.7. Zero-Dependency i18n

Engine (

`translations.ts`)

Pure TypeScript localization dictionary:

- **4 Supported Languages:** English (`en`), Hindi (`hi`), Assamese (`as`), Bengali (`bn`).
- **150+ Keys per Language:** Translates UI text, MoRTH crash stats, vehicle specs, weather alerts, and 111 city names with zero library runtime overhead.

4. Dual-Solve Routing Architecture & API Integration

The core innovation of RaahSetu is its **Dual-Solve Algorithm**. The frontend simultaneously evaluates two independent paths for any given origin, destination, vehicle classification, and cargo sensitivity:

Routing Criterion	Fastest Route (Conventional Nav)	RaahSetu Risk-Aware Recommended Corridor
Objective Function	<code>min(Total Travel Time)</code>	<code>min(Time + λ × Risk_Exposure + Bridge_Penalty + Slope_Penalty)</code>
Slope Consideration	Ignores elevation gradients; frequently routes over steep 14%+ ghat shortcuts.	Penalizes steep ascents/descents to avoid heavy truck brake overheating and roll-aways.
Bridge Constraints	Unaware of axle load limits on temporary timber and Bailey bridges.	Strictly checks vehicle Gross Vehicle Weight (GVW) against bridge class capacity.
Commodity Sensitivity	Treats volatile petroleum fuel tankers the same as light passenger cars.	Prioritizes low vibration for medical supplies and avoids landslide zones for fuel tankers.
Visual Map Display	Amber / Red dashed polyline.	Glowing Emerald solid polyline with directional arrows.
Solvers Available	Local Dijkstra (<code>routeData.ts</code>) + FastAPI A* (<code>/api/v1/routes/compare</code>)	Local Dijkstra (<code>routeData.ts</code>) + FastAPI A* (<code>/api/v1/routes/compare</code>)

5. End-to-End User Experience Lifecycle

[1. Landing Page / Ground Reality] —> [2. Authentication & Verification] —> [3. Vehicle & Cargo Manifest] • MoRTH / NCRB Census Data • Sign In or Register • 1-Click Archetype Presets • Problem Statement Analysis • Procedural Canvas Captcha • Gross Vehicle Weight (GVW) • Interactive 3D Constellation • Two-Factor Email OTP • Bailey Bridge & Tunnel Checks | ▼ [6. Offline Resiliency / Dead Zones] <— [5. Interactive 2D/3D Navigation] <— [4. Route Dispatcher Workspace] • LocalStorage Hazard Report Queue • Leaflet Multi-Layer 2D Map • Select Origin & Destination • Local Client-Side Dijkstra Fallback • Three.js WebGL 3D Terrain • Live A* Comparison Telemetry • Offline SOS Distress Dispatch • Hands-Free Multilingual TTS Audio • Real-time Travel & Risk Metrics

6. Frontend File & Module Directory

File Path	Size	Module Purpose & Core Exports
src/App.tsx	121 KB	Root application controller, view state machine, theme engine, telemetry header, and modal coordinators.
src/VehicleDetailsPage.tsx	63 KB	Post-login vehicle manifest, GVW Bailey bridge gauge, dimensional clearances, and pre-departure safety checklist.
src/AuthPage.tsx	28 KB	Driver authentication, sign-up, HTML5 canvas captcha generator, and two-factor email OTP verification.
src/IntroPage.tsx	26 KB	Ground reality showcase, MoRTH official accident statistics, GSI landslide records, and SIH PS analysis.
src/RealMapLeaflet.tsx	13 KB	Interactive 2D Leaflet map, OpenStreetMap tiles, custom SVG pinpoint markers, and dual route polylines.
src/TerrainMap.tsx	15 KB	Three.js WebGL 3D elevation mesh viewer, procedural altitude shader, and OrbitControls camera rig.
src/routeData.ts	64 KB	111 Northeast cities, 378 directed road corridor edges, and offline client-side Dijkstra solver.
src/translations.ts	216 KB	Zero-dependency multilingual dictionary covering English, Hindi, Assamese, and Bengali (150+ keys each).
src/voiceRecognition.ts	17 KB	Web Speech API implementation with phonetic fuzzy city matching and multilingual turn-by-turn TTS audio.
src/api.ts	8 KB	Type-safe REST client for FastAPI endpoints (/routes/compare , /network , /terrain , /field-reports).
src/offlineQueue.ts	2.5 KB	LocalStorage transaction queue for field reports and SOS alerts during high-altitude connectivity blackouts.
vite.config.ts	0.5 KB	Vite 7 build configuration with React plugin, Tailwind CSS v4 compiler, and reverse proxy rules.

7. Verification & Production Deployment

The RaahSetu frontend has been verified across Chrome, Edge, and mobile viewports. It passes 100% strict TypeScript checks (`tsc -b`) and builds in **8.39 seconds** into optimized production chunks:

- dist/index.html (1.04 kB | gzip: 0.56 kB)
- dist/assets/index-[hash].css (118.46 kB | gzip: 24.32 kB)
- dist/assets/index-[hash].js (1,672.85 kB | gzip: 459.07 kB)

The bundle is hosted directly by the FastAPI backend on port `8000` , ensuring a unified single-link delivery for drivers and evaluators: `http://127.0.0.1:8000/` .